

Table 1: Complete ablation findings with selected task metrics and LM parameters. Analogy is reported with Accuracy; Morphology, NER, POS, and Sentiment are reported with F-measure. A legacy status marks sequence-labeling artifacts without accompanying Precision and Recall fields.

| Task    | Method         | ID  | Win. | Slide | Top | Skip | Stem | Prune | $w_{\text{like}}$ | $w_{\text{prior}}$ | Penalty            | Metric   | Score    | Status  |
|---------|----------------|-----|------|-------|-----|------|------|-------|-------------------|--------------------|--------------------|----------|----------|---------|
| ANALOGY | frequent-ngram | v1  | 2    | 2     | 1   | 2    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.016695 | current |
| ANALOGY | frequent-ngram | v2  | 2    | 2     | 3   | 2    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.017340 | current |
| ANALOGY | frequent-ngram | v3  | 2    | 2     | 5   | 2    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.016624 | current |
| ANALOGY | frequent-ngram | v4  | 3    | 3     | 1   | 3    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.017555 | current |
| ANALOGY | frequent-ngram | v5  | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.017197 | current |
| ANALOGY | frequent-ngram | v6  | 3    | 3     | 5   | 3    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.016409 | current |
| ANALOGY | frequent-ngram | v7  | 4    | 4     | 1   | 4    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.017484 | current |
| ANALOGY | frequent-ngram | v8  | 4    | 4     | 3   | 4    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.016624 | current |
| ANALOGY | frequent-ngram | v9  | 4    | 4     | 5   | 4    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.016122 | current |
| ANALOGY | lm-lemma       | v1  | 2    | 2     | 1   | 2    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.030883 | current |
| ANALOGY | lm-lemma       | v2  | 2    | 2     | 3   | 2    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.029880 | current |
| ANALOGY | lm-lemma       | v3  | 2    | 2     | 5   | 2    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.029951 | current |
| ANALOGY | lm-lemma       | v4  | 3    | 3     | 1   | 3    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.030381 | current |
| ANALOGY | lm-lemma       | v5  | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.031743 | current |
| ANALOGY | lm-lemma       | v6  | 3    | 3     | 5   | 3    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.031026 | current |
| ANALOGY | lm-lemma       | v7  | 4    | 4     | 1   | 4    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.030739 | current |
| ANALOGY | lm-lemma       | v8  | 4    | 4     | 3   | 4    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.030453 | current |
| ANALOGY | lm-lemma       | v9  | 4    | 4     | 5   | 4    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.030883 | current |
| ANALOGY | lm-lemma       | v10 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | none               | Accuracy | 0.031241 | current |
| ANALOGY | lm-lemma       | v11 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | inverse_parts      | Accuracy | 0.031743 | current |
| ANALOGY | lm-lemma       | v12 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | inverse_sqrt_parts | Accuracy | 0.031528 | current |
| ANALOGY | lm-lemma       | v13 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | inverse_log_parts  | Accuracy | 0.032531 | current |
| ANALOGY | lm-lemma       | v14 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | none               | Accuracy | 0.031599 | current |
| ANALOGY | lm-lemma       | v15 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | inverse_parts      | Accuracy | 0.031743 | current |
| ANALOGY | lm-lemma       | v16 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | inverse_sqrt_parts | Accuracy | 0.030668 | current |
| ANALOGY | lm-lemma       | v17 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | inverse_log_parts  | Accuracy | 0.032388 | current |
| ANALOGY | lm-lemma       | v18 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | inverse_parts      | Accuracy | 0.030525 | current |
| ANALOGY | lm-lemma       | v19 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | inverse_sqrt_parts | Accuracy | 0.032029 | current |
| ANALOGY | lm-lemma       | v20 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | inverse_log_parts  | Accuracy | 0.030883 | current |
| ANALOGY | lm-lemma       | v21 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | none               | Accuracy | 0.031456 | current |
| ANALOGY | lm-lemma       | v22 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | inverse_parts      | Accuracy | 0.031169 | current |
| ANALOGY | lm-lemma       | v23 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | inverse_sqrt_parts | Accuracy | 0.032173 | current |
| ANALOGY | lm-lemma       | v24 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | inverse_log_parts  | Accuracy | 0.031098 | current |
| ANALOGY | lm-lemma       | v25 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | none               | Accuracy | 0.031098 | current |
| ANALOGY | lm-lemma       | v26 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | inverse_parts      | Accuracy | 0.032602 | current |
| ANALOGY | lm-lemma       | v27 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | inverse_sqrt_parts | Accuracy | 0.030811 | current |
| ANALOGY | lm-lemma       | v28 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | inverse_log_parts  | Accuracy | 0.031743 | current |
| ANALOGY | lm-rank        | v1  | 2    | 2     | 1   | 2    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.000143 | current |
| ANALOGY | lm-rank        | v2  | 2    | 2     | 3   | 2    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.000645 | current |
| ANALOGY | lm-rank        | v3  | 2    | 2     | 5   | 2    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.001863 | current |
| ANALOGY | lm-rank        | v4  | 3    | 3     | 1   | 3    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.000143 | current |
| ANALOGY | lm-rank        | v5  | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.000215 | current |
| ANALOGY | lm-rank        | v6  | 3    | 3     | 5   | 3    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.001505 | current |
| ANALOGY | lm-rank        | v7  | 4    | 4     | 1   | 4    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.000215 | current |
| ANALOGY | lm-rank        | v8  | 4    | 4     | 3   | 4    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.000860 | current |
| ANALOGY | lm-rank        | v9  | 4    | 4     | 5   | 4    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.000788 | current |
| ANALOGY | lm-rank        | v10 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | none               | Accuracy | 0.000430 | current |
| ANALOGY | lm-rank        | v11 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | inverse_parts      | Accuracy | 0.000358 | current |
| ANALOGY | lm-rank        | v12 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | inverse_sqrt_parts | Accuracy | 0.000502 | current |
| ANALOGY | lm-rank        | v13 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | inverse_log_parts  | Accuracy | 0.000573 | current |
| ANALOGY | lm-rank        | v14 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | none               | Accuracy | 0.000215 | current |
| ANALOGY | lm-rank        | v15 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | inverse_parts      | Accuracy | 0.000430 | current |
| ANALOGY | lm-rank        | v16 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | inverse_sqrt_parts | Accuracy | 0.000430 | current |
| ANALOGY | lm-rank        | v17 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | inverse_log_parts  | Accuracy | 0.000430 | current |
| ANALOGY | lm-rank        | v18 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | inverse_parts      | Accuracy | 0.000358 | current |
| ANALOGY | lm-rank        | v19 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | inverse_sqrt_parts | Accuracy | 0.000358 | current |
| ANALOGY | lm-rank        | v20 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | inverse_log_parts  | Accuracy | 0.000358 | current |
| ANALOGY | lm-rank        | v21 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | none               | Accuracy | 0.000645 | current |
| ANALOGY | lm-rank        | v22 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | inverse_parts      | Accuracy | 0.000645 | current |
| ANALOGY | lm-rank        | v23 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | inverse_sqrt_parts | Accuracy | 0.000645 | current |

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Table 1 — continued

| Task       | Method         | ID  | Win. | Slide | Top | Skip | Stem | Prune | $w_{\text{like}}$ | $w_{\text{prior}}$ | Penalty            | Metric   | Score    | Status  |
|------------|----------------|-----|------|-------|-----|------|------|-------|-------------------|--------------------|--------------------|----------|----------|---------|
| ANALOGY    | lm-rank        | v24 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | inverse_log_parts  | Accuracy | 0.000788 | current |
| ANALOGY    | lm-rank        | v25 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | none               | Accuracy | 0.000645 | current |
| ANALOGY    | lm-rank        | v26 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | inverse_parts      | Accuracy | 0.000717 | current |
| ANALOGY    | lm-rank        | v27 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | inverse_sqrt_parts | Accuracy | 0.000860 | current |
| ANALOGY    | lm-rank        | v28 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | inverse_log_parts  | Accuracy | 0.000860 | current |
| ANALOGY    | lm-skip        | v1  | 2    | 2     | 1   | 2    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.000143 | current |
| ANALOGY    | lm-skip        | v2  | 2    | 2     | 3   | 2    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.020063 | current |
| ANALOGY    | lm-skip        | v3  | 2    | 2     | 5   | 2    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.016194 | current |
| ANALOGY    | lm-skip        | v4  | 3    | 3     | 1   | 3    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.000215 | current |
| ANALOGY    | lm-skip        | v5  | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.020565 | current |
| ANALOGY    | lm-skip        | v6  | 3    | 3     | 5   | 3    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.015262 | current |
| ANALOGY    | lm-skip        | v7  | 4    | 4     | 1   | 4    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.000143 | current |
| ANALOGY    | lm-skip        | v8  | 4    | 4     | 3   | 4    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.020780 | current |
| ANALOGY    | lm-skip        | v9  | 4    | 4     | 5   | 4    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.016265 | current |
| ANALOGY    | lm-subword     | v1  | 2    | 2     | 1   | 2    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.000215 | current |
| ANALOGY    | lm-subword     | v2  | 2    | 2     | 3   | 2    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.034752 | current |
| ANALOGY    | lm-subword     | v3  | 2    | 2     | 5   | 2    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.030166 | current |
| ANALOGY    | lm-subword     | v4  | 3    | 3     | 1   | 3    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.000215 | current |
| ANALOGY    | lm-subword     | v5  | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.035397 | current |
| ANALOGY    | lm-subword     | v6  | 3    | 3     | 5   | 3    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.028805 | current |
| ANALOGY    | lm-subword     | v7  | 4    | 4     | 1   | 4    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.000287 | current |
| ANALOGY    | lm-subword     | v8  | 4    | 4     | 3   | 4    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.037762 | current |
| ANALOGY    | lm-subword     | v9  | 4    | 4     | 5   | 4    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.029235 | current |
| ANALOGY    | lm-subword     | v10 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.033964 | current |
| ANALOGY    | lm-subword     | v11 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.037117 | current |
| ANALOGY    | lm-subword     | v12 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.035899 | current |
| ANALOGY    | lm-syllable    | v1  | 2    | 2     | 1   | 2    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.000000 | current |
| ANALOGY    | lm-syllable    | v2  | 2    | 2     | 3   | 2    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.000788 | current |
| ANALOGY    | lm-syllable    | v3  | 2    | 2     | 5   | 2    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.002078 | current |
| ANALOGY    | lm-syllable    | v4  | 3    | 3     | 1   | 3    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.000072 | current |
| ANALOGY    | lm-syllable    | v5  | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.000788 | current |
| ANALOGY    | lm-syllable    | v6  | 3    | 3     | 5   | 3    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.002078 | current |
| ANALOGY    | lm-syllable    | v7  | 4    | 4     | 1   | 4    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.000000 | current |
| ANALOGY    | lm-syllable    | v8  | 4    | 4     | 3   | 4    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.001003 | current |
| ANALOGY    | lm-syllable    | v9  | 4    | 4     | 5   | 4    | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.001863 | current |
| ANALOGY    | lm-word        | v1  | 10   | 7     | 10  | 20   | 7    | 100   | 0.15              | 0.85               | none               | Accuracy | 0.000143 | current |
| MORPHOLOGY | frequent-ngram | v1  | 2    | 2     | 1   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score  | 0.345762 | current |
| MORPHOLOGY | frequent-ngram | v2  | 2    | 2     | 3   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score  | 0.345762 | current |
| MORPHOLOGY | frequent-ngram | v3  | 2    | 2     | 5   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score  | 0.345762 | current |
| MORPHOLOGY | frequent-ngram | v4  | 3    | 3     | 1   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score  | 0.345762 | current |
| MORPHOLOGY | frequent-ngram | v5  | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score  | 0.345762 | current |
| MORPHOLOGY | frequent-ngram | v6  | 3    | 3     | 5   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score  | 0.345762 | current |
| MORPHOLOGY | frequent-ngram | v7  | 4    | 4     | 1   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score  | 0.345762 | current |
| MORPHOLOGY | frequent-ngram | v8  | 4    | 4     | 3   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score  | 0.345762 | current |
| MORPHOLOGY | frequent-ngram | v9  | 4    | 4     | 5   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score  | 0.345762 | current |
| MORPHOLOGY | lm-lemma       | v1  | 2    | 2     | 1   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score  | 0.475877 | current |
| MORPHOLOGY | lm-lemma       | v2  | 2    | 2     | 3   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score  | 0.475877 | current |
| MORPHOLOGY | lm-lemma       | v3  | 2    | 2     | 5   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score  | 0.475877 | current |
| MORPHOLOGY | lm-lemma       | v4  | 3    | 3     | 1   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score  | 0.475877 | current |
| MORPHOLOGY | lm-lemma       | v5  | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score  | 0.475877 | current |
| MORPHOLOGY | lm-lemma       | v6  | 3    | 3     | 5   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score  | 0.475877 | current |
| MORPHOLOGY | lm-lemma       | v7  | 4    | 4     | 1   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score  | 0.475877 | current |
| MORPHOLOGY | lm-lemma       | v8  | 4    | 4     | 3   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score  | 0.475877 | current |
| MORPHOLOGY | lm-lemma       | v9  | 4    | 4     | 5   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score  | 0.475877 | current |
| MORPHOLOGY | lm-lemma       | v10 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | none               | F-Score  | 0.475877 | current |
| MORPHOLOGY | lm-lemma       | v11 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | inverse_parts      | F-Score  | 0.475877 | current |
| MORPHOLOGY | lm-lemma       | v12 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | inverse_sqrt_parts | F-Score  | 0.475877 | current |
| MORPHOLOGY | lm-lemma       | v13 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | inverse_log_parts  | F-Score  | 0.475877 | current |
| MORPHOLOGY | lm-lemma       | v14 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | none               | F-Score  | 0.475877 | current |
| MORPHOLOGY | lm-lemma       | v15 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | inverse_parts      | F-Score  | 0.475877 | current |
| MORPHOLOGY | lm-lemma       | v16 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | inverse_sqrt_parts | F-Score  | 0.475877 | current |
| MORPHOLOGY | lm-lemma       | v17 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | inverse_log_parts  | F-Score  | 0.475877 | current |
| MORPHOLOGY | lm-lemma       | v18 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | inverse_parts      | F-Score  | 0.475877 | current |
| MORPHOLOGY | lm-lemma       | v19 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | inverse_sqrt_parts | F-Score  | 0.475877 | current |

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Table 1 — continued

| Task       | Method      | ID  | Win. | Slide | Top | Skip | Stem | Prune | $w_{\text{like}}$ | $w_{\text{prior}}$ | Penalty            | Metric  | Score    | Status  |
|------------|-------------|-----|------|-------|-----|------|------|-------|-------------------|--------------------|--------------------|---------|----------|---------|
| MORPHOLOGY | lm-lemma    | v20 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | inverse_log_parts  | F-Score | 0.475877 | current |
| MORPHOLOGY | lm-lemma    | v21 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | none               | F-Score | 0.475877 | current |
| MORPHOLOGY | lm-lemma    | v22 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | inverse_parts      | F-Score | 0.475877 | current |
| MORPHOLOGY | lm-lemma    | v23 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | inverse_sqrt_parts | F-Score | 0.475877 | current |
| MORPHOLOGY | lm-lemma    | v24 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | inverse_log_parts  | F-Score | 0.475877 | current |
| MORPHOLOGY | lm-lemma    | v25 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | none               | F-Score | 0.475877 | current |
| MORPHOLOGY | lm-lemma    | v26 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | inverse_parts      | F-Score | 0.475877 | current |
| MORPHOLOGY | lm-lemma    | v27 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | inverse_sqrt_parts | F-Score | 0.475877 | current |
| MORPHOLOGY | lm-lemma    | v28 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | inverse_log_parts  | F-Score | 0.475877 | current |
| MORPHOLOGY | lm-rank     | v1  | 2    | 2     | 1   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.486911 | current |
| MORPHOLOGY | lm-rank     | v2  | 2    | 2     | 3   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.439903 | current |
| MORPHOLOGY | lm-rank     | v3  | 2    | 2     | 5   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.397296 | current |
| MORPHOLOGY | lm-rank     | v4  | 3    | 3     | 1   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.486911 | current |
| MORPHOLOGY | lm-rank     | v5  | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.417918 | current |
| MORPHOLOGY | lm-rank     | v6  | 3    | 3     | 5   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.383820 | current |
| MORPHOLOGY | lm-rank     | v7  | 4    | 4     | 1   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.486911 | current |
| MORPHOLOGY | lm-rank     | v8  | 4    | 4     | 3   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.370133 | current |
| MORPHOLOGY | lm-rank     | v9  | 4    | 4     | 5   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.324811 | current |
| MORPHOLOGY | lm-rank     | v10 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | none               | F-Score | 0.486911 | current |
| MORPHOLOGY | lm-rank     | v11 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | inverse_parts      | F-Score | 0.493722 | current |
| MORPHOLOGY | lm-rank     | v12 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | inverse_sqrt_parts | F-Score | 0.492223 | current |
| MORPHOLOGY | lm-rank     | v13 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | inverse_log_parts  | F-Score | 0.492889 | current |
| MORPHOLOGY | lm-rank     | v14 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | none               | F-Score | 0.429443 | current |
| MORPHOLOGY | lm-rank     | v15 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | inverse_parts      | F-Score | 0.425090 | current |
| MORPHOLOGY | lm-rank     | v16 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | inverse_sqrt_parts | F-Score | 0.426740 | current |
| MORPHOLOGY | lm-rank     | v17 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | inverse_log_parts  | F-Score | 0.428633 | current |
| MORPHOLOGY | lm-rank     | v18 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | inverse_parts      | F-Score | 0.416921 | current |
| MORPHOLOGY | lm-rank     | v19 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | inverse_sqrt_parts | F-Score | 0.417018 | current |
| MORPHOLOGY | lm-rank     | v20 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | inverse_log_parts  | F-Score | 0.417679 | current |
| MORPHOLOGY | lm-rank     | v21 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | none               | F-Score | 0.413978 | current |
| MORPHOLOGY | lm-rank     | v22 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | inverse_parts      | F-Score | 0.411829 | current |
| MORPHOLOGY | lm-rank     | v23 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | inverse_sqrt_parts | F-Score | 0.412023 | current |
| MORPHOLOGY | lm-rank     | v24 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | inverse_log_parts  | F-Score | 0.414004 | current |
| MORPHOLOGY | lm-rank     | v25 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | none               | F-Score | 0.415739 | current |
| MORPHOLOGY | lm-rank     | v26 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | inverse_parts      | F-Score | 0.415739 | current |
| MORPHOLOGY | lm-rank     | v27 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | inverse_sqrt_parts | F-Score | 0.415739 | current |
| MORPHOLOGY | lm-rank     | v28 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | inverse_log_parts  | F-Score | 0.415739 | current |
| MORPHOLOGY | lm-skip     | v1  | 2    | 2     | 1   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.538576 | current |
| MORPHOLOGY | lm-skip     | v2  | 2    | 2     | 3   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.247958 | current |
| MORPHOLOGY | lm-skip     | v3  | 2    | 2     | 5   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.182867 | current |
| MORPHOLOGY | lm-skip     | v4  | 3    | 3     | 1   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.538576 | current |
| MORPHOLOGY | lm-skip     | v5  | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.247958 | current |
| MORPHOLOGY | lm-skip     | v6  | 3    | 3     | 5   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.182867 | current |
| MORPHOLOGY | lm-skip     | v7  | 4    | 4     | 1   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.538576 | current |
| MORPHOLOGY | lm-skip     | v8  | 4    | 4     | 3   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.247958 | current |
| MORPHOLOGY | lm-skip     | v9  | 4    | 4     | 5   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.182867 | current |
| MORPHOLOGY | lm-subword  | v1  | 2    | 2     | 1   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.357329 | current |
| MORPHOLOGY | lm-subword  | v2  | 2    | 2     | 3   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.222423 | current |
| MORPHOLOGY | lm-subword  | v3  | 2    | 2     | 5   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.198302 | current |
| MORPHOLOGY | lm-subword  | v4  | 3    | 3     | 1   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.357329 | current |
| MORPHOLOGY | lm-subword  | v5  | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.224409 | current |
| MORPHOLOGY | lm-subword  | v6  | 3    | 3     | 5   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.191012 | current |
| MORPHOLOGY | lm-subword  | v7  | 4    | 4     | 1   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.357329 | current |
| MORPHOLOGY | lm-subword  | v8  | 4    | 4     | 3   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.226115 | current |
| MORPHOLOGY | lm-subword  | v9  | 4    | 4     | 5   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.199732 | current |
| MORPHOLOGY | lm-subword  | v10 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.230668 | current |
| MORPHOLOGY | lm-subword  | v11 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.226355 | current |
| MORPHOLOGY | lm-subword  | v12 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.234597 | current |
| MORPHOLOGY | lm-syllable | v1  | 2    | 2     | 1   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.358150 | current |
| MORPHOLOGY | lm-syllable | v2  | 2    | 2     | 3   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.212726 | current |
| MORPHOLOGY | lm-syllable | v3  | 2    | 2     | 5   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.218891 | current |
| MORPHOLOGY | lm-syllable | v4  | 3    | 3     | 1   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.358150 | current |
| MORPHOLOGY | lm-syllable | v5  | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.212196 | current |
| MORPHOLOGY | lm-syllable | v6  | 3    | 3     | 5   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.217637 | current |

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Table 1 — continued

| Task       | Method         | ID  | Win. | Slide | Top | Skip | Stem | Prune | $w_{\text{like}}$ | $w_{\text{prior}}$ | Penalty            | Metric  | Score    | Status  |
|------------|----------------|-----|------|-------|-----|------|------|-------|-------------------|--------------------|--------------------|---------|----------|---------|
| MORPHOLOGY | lm-syllable    | v7  | 4    | 4     | 1   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.358150 | current |
| MORPHOLOGY | lm-syllable    | v8  | 4    | 4     | 3   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.211134 | current |
| MORPHOLOGY | lm-syllable    | v9  | 4    | 4     | 5   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.221651 | current |
| MORPHOLOGY | lm-word        | v1  | 10   | 7     | 10  | 20   | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.345762 | current |
| NER        | frequent-ngram | v1  | 2    | 2     | 1   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.402539 | current |
| NER        | frequent-ngram | v2  | 2    | 2     | 3   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.389274 | current |
| NER        | frequent-ngram | v3  | 2    | 2     | 5   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.400716 | current |
| NER        | frequent-ngram | v4  | 3    | 3     | 1   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.406839 | current |
| NER        | frequent-ngram | v5  | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.407218 | current |
| NER        | frequent-ngram | v6  | 3    | 3     | 5   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.394611 | current |
| NER        | frequent-ngram | v7  | 4    | 4     | 1   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.406508 | current |
| NER        | frequent-ngram | v8  | 4    | 4     | 3   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.391313 | current |
| NER        | frequent-ngram | v9  | 4    | 4     | 5   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.404484 | current |
| NER        | lm-lemma       | v1  | 2    | 2     | 1   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.631511 | current |
| NER        | lm-lemma       | v2  | 2    | 2     | 3   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.646740 | current |
| NER        | lm-lemma       | v3  | 2    | 2     | 5   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.651788 | current |
| NER        | lm-lemma       | v4  | 3    | 3     | 1   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.656447 | current |
| NER        | lm-lemma       | v5  | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.654494 | current |
| NER        | lm-lemma       | v6  | 3    | 3     | 5   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.656965 | current |
| NER        | lm-lemma       | v7  | 4    | 4     | 1   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.651899 | current |
| NER        | lm-lemma       | v8  | 4    | 4     | 3   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.655353 | current |
| NER        | lm-lemma       | v9  | 4    | 4     | 5   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.652893 | current |
| NER        | lm-lemma       | v10 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | none               | F-Score | 0.645930 | current |
| NER        | lm-lemma       | v11 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | inverse_parts      | F-Score | 0.660480 | current |
| NER        | lm-lemma       | v12 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | inverse_sqrt_parts | F-Score | 0.643570 | current |
| NER        | lm-lemma       | v13 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | inverse_log_parts  | F-Score | 0.654287 | current |
| NER        | lm-lemma       | v14 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | none               | F-Score | 0.645835 | current |
| NER        | lm-lemma       | v15 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | inverse_parts      | F-Score | 0.645812 | current |
| NER        | lm-lemma       | v16 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | inverse_sqrt_parts | F-Score | 0.642653 | current |
| NER        | lm-lemma       | v17 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | inverse_log_parts  | F-Score | 0.651667 | current |
| NER        | lm-lemma       | v18 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | inverse_parts      | F-Score | 0.643738 | current |
| NER        | lm-lemma       | v19 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | inverse_sqrt_parts | F-Score | 0.649443 | current |
| NER        | lm-lemma       | v20 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | inverse_log_parts  | F-Score | 0.638591 | current |
| NER        | lm-lemma       | v21 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | none               | F-Score | 0.652414 | current |
| NER        | lm-lemma       | v22 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | inverse_parts      | F-Score | 0.636522 | current |
| NER        | lm-lemma       | v23 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | inverse_sqrt_parts | F-Score | 0.631716 | current |
| NER        | lm-lemma       | v24 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | inverse_log_parts  | F-Score | 0.656470 | current |
| NER        | lm-lemma       | v25 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | none               | F-Score | 0.647740 | current |
| NER        | lm-lemma       | v26 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | inverse_parts      | F-Score | 0.651672 | current |
| NER        | lm-lemma       | v27 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | inverse_sqrt_parts | F-Score | 0.649621 | current |
| NER        | lm-lemma       | v28 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | inverse_log_parts  | F-Score | 0.648994 | current |
| NER        | lm-rank        | v1  | 2    | 2     | 1   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.526664 | current |
| NER        | lm-rank        | v2  | 2    | 2     | 3   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.493389 | current |
| NER        | lm-rank        | v3  | 2    | 2     | 5   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.512705 | current |
| NER        | lm-rank        | v4  | 3    | 3     | 1   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.515936 | current |
| NER        | lm-rank        | v5  | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.484380 | current |
| NER        | lm-rank        | v6  | 3    | 3     | 5   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.512234 | current |
| NER        | lm-rank        | v7  | 4    | 4     | 1   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.511715 | current |
| NER        | lm-rank        | v8  | 4    | 4     | 3   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.497677 | current |
| NER        | lm-rank        | v9  | 4    | 4     | 5   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.509056 | current |
| NER        | lm-rank        | v10 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | none               | F-Score | 0.502652 | current |
| NER        | lm-rank        | v11 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | inverse_parts      | F-Score | 0.509035 | current |
| NER        | lm-rank        | v12 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | inverse_sqrt_parts | F-Score | 0.478306 | current |
| NER        | lm-rank        | v13 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | inverse_log_parts  | F-Score | 0.501479 | current |
| NER        | lm-rank        | v14 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | none               | F-Score | 0.491424 | current |
| NER        | lm-rank        | v15 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | inverse_parts      | F-Score | 0.506898 | current |
| NER        | lm-rank        | v16 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | inverse_sqrt_parts | F-Score | 0.515854 | current |
| NER        | lm-rank        | v17 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | inverse_log_parts  | F-Score | 0.491710 | current |
| NER        | lm-rank        | v18 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | inverse_parts      | F-Score | 0.488015 | current |
| NER        | lm-rank        | v19 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | inverse_sqrt_parts | F-Score | 0.486725 | current |
| NER        | lm-rank        | v20 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | inverse_log_parts  | F-Score | 0.478386 | current |
| NER        | lm-rank        | v21 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | none               | F-Score | 0.476779 | current |
| NER        | lm-rank        | v22 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | inverse_parts      | F-Score | 0.488815 | current |
| NER        | lm-rank        | v23 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | inverse_sqrt_parts | F-Score | 0.477688 | current |

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Table 1 — continued

| Task | Method         | ID  | Win. | Slide | Top | Skip | Stem | Prune | $w_{\text{like}}$ | $w_{\text{prior}}$ | Penalty            | Metric  | Score    | Status  |
|------|----------------|-----|------|-------|-----|------|------|-------|-------------------|--------------------|--------------------|---------|----------|---------|
| NER  | lm-rank        | v24 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | inverse_log_parts  | F-Score | 0.482752 | current |
| NER  | lm-rank        | v25 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | none               | F-Score | 0.491259 | current |
| NER  | lm-rank        | v26 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | inverse_parts      | F-Score | 0.498209 | current |
| NER  | lm-rank        | v27 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | inverse_sqrt_parts | F-Score | 0.517694 | current |
| NER  | lm-rank        | v28 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | inverse_log_parts  | F-Score | 0.502094 | current |
| NER  | lm-skip        | v1  | 2    | 2     | 1   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.664319 | current |
| NER  | lm-skip        | v2  | 2    | 2     | 3   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.565077 | current |
| NER  | lm-skip        | v3  | 2    | 2     | 5   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.524024 | current |
| NER  | lm-skip        | v4  | 3    | 3     | 1   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.658627 | current |
| NER  | lm-skip        | v5  | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.584255 | current |
| NER  | lm-skip        | v6  | 3    | 3     | 5   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.537100 | current |
| NER  | lm-skip        | v7  | 4    | 4     | 1   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.669862 | current |
| NER  | lm-skip        | v8  | 4    | 4     | 3   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.573471 | current |
| NER  | lm-skip        | v9  | 4    | 4     | 5   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.542219 | current |
| NER  | lm-subword     | v1  | 2    | 2     | 1   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.591843 | current |
| NER  | lm-subword     | v2  | 2    | 2     | 3   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.571039 | current |
| NER  | lm-subword     | v3  | 2    | 2     | 5   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.520242 | current |
| NER  | lm-subword     | v4  | 3    | 3     | 1   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.533782 | current |
| NER  | lm-subword     | v5  | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.559435 | current |
| NER  | lm-subword     | v6  | 3    | 3     | 5   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.557017 | current |
| NER  | lm-subword     | v7  | 4    | 4     | 1   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.545936 | current |
| NER  | lm-subword     | v8  | 4    | 4     | 3   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.560875 | current |
| NER  | lm-subword     | v9  | 4    | 4     | 5   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.532700 | current |
| NER  | lm-subword     | v10 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.561357 | current |
| NER  | lm-subword     | v11 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.552080 | current |
| NER  | lm-subword     | v12 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.571702 | current |
| NER  | lm-syllable    | v1  | 2    | 2     | 1   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.675507 | current |
| NER  | lm-syllable    | v2  | 2    | 2     | 3   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.540801 | current |
| NER  | lm-syllable    | v3  | 2    | 2     | 5   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.554074 | current |
| NER  | lm-syllable    | v4  | 3    | 3     | 1   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.679995 | current |
| NER  | lm-syllable    | v5  | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.531075 | current |
| NER  | lm-syllable    | v6  | 3    | 3     | 5   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.557449 | current |
| NER  | lm-syllable    | v7  | 4    | 4     | 1   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.682859 | current |
| NER  | lm-syllable    | v8  | 4    | 4     | 3   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.536667 | current |
| NER  | lm-syllable    | v9  | 4    | 4     | 5   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.549905 | current |
| NER  | lm-word        | v1  | 10   | 7     | 10  | 20   | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.126547 | current |
| POS  | frequent-ngram | v1  | 2    | 2     | 1   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.477612 | current |
| POS  | frequent-ngram | v2  | 2    | 2     | 3   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.481695 | current |
| POS  | frequent-ngram | v3  | 2    | 2     | 5   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.462311 | current |
| POS  | frequent-ngram | v4  | 3    | 3     | 1   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.473343 | current |
| POS  | frequent-ngram | v5  | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.477751 | current |
| POS  | frequent-ngram | v6  | 3    | 3     | 5   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.462524 | current |
| POS  | frequent-ngram | v7  | 4    | 4     | 1   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.473225 | current |
| POS  | frequent-ngram | v8  | 4    | 4     | 3   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.460453 | current |
| POS  | frequent-ngram | v9  | 4    | 4     | 5   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.471494 | current |
| POS  | lm-lemma       | v1  | 2    | 2     | 1   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.627158 | current |
| POS  | lm-lemma       | v2  | 2    | 2     | 3   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.627966 | current |
| POS  | lm-lemma       | v3  | 2    | 2     | 5   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.634812 | current |
| POS  | lm-lemma       | v4  | 3    | 3     | 1   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.623650 | current |
| POS  | lm-lemma       | v5  | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.614353 | current |
| POS  | lm-lemma       | v6  | 3    | 3     | 5   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.618727 | current |
| POS  | lm-lemma       | v7  | 4    | 4     | 1   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.630793 | current |
| POS  | lm-lemma       | v8  | 4    | 4     | 3   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.645612 | current |
| POS  | lm-lemma       | v9  | 4    | 4     | 5   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.616657 | current |
| POS  | lm-lemma       | v10 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | none               | F-Score | 0.621752 | current |
| POS  | lm-lemma       | v11 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | inverse_parts      | F-Score | 0.627611 | current |
| POS  | lm-lemma       | v12 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | inverse_sqrt_parts | F-Score | 0.622077 | current |
| POS  | lm-lemma       | v13 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | inverse_log_parts  | F-Score | 0.621367 | current |
| POS  | lm-lemma       | v14 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | none               | F-Score | 0.617064 | current |
| POS  | lm-lemma       | v15 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | inverse_parts      | F-Score | 0.618803 | current |
| POS  | lm-lemma       | v16 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | inverse_sqrt_parts | F-Score | 0.634761 | current |
| POS  | lm-lemma       | v17 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | inverse_log_parts  | F-Score | 0.635365 | current |
| POS  | lm-lemma       | v18 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | inverse_parts      | F-Score | 0.614236 | current |
| POS  | lm-lemma       | v19 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | inverse_sqrt_parts | F-Score | 0.626998 | current |

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Table 1 — continued

| Task | Method      | ID  | Win. | Slide | Top | Skip | Stem | Prune | $w_{\text{like}}$ | $w_{\text{prior}}$ | Penalty            | Metric  | Score    | Status  |
|------|-------------|-----|------|-------|-----|------|------|-------|-------------------|--------------------|--------------------|---------|----------|---------|
| POS  | lm-lemma    | v20 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | inverse_log_parts  | F-Score | 0.625807 | current |
| POS  | lm-lemma    | v21 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | none               | F-Score | 0.622750 | current |
| POS  | lm-lemma    | v22 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | inverse_parts      | F-Score | 0.631396 | current |
| POS  | lm-lemma    | v23 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | inverse_sqrt_parts | F-Score | 0.630219 | current |
| POS  | lm-lemma    | v24 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | inverse_log_parts  | F-Score | 0.623450 | current |
| POS  | lm-lemma    | v25 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | none               | F-Score | 0.621703 | current |
| POS  | lm-lemma    | v26 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | inverse_parts      | F-Score | 0.630275 | current |
| POS  | lm-lemma    | v27 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | inverse_sqrt_parts | F-Score | 0.623565 | current |
| POS  | lm-lemma    | v28 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | inverse_log_parts  | F-Score | 0.629394 | current |
| POS  | lm-rank     | v1  | 2    | 2     | 1   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.625917 | current |
| POS  | lm-rank     | v2  | 2    | 2     | 3   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.558886 | current |
| POS  | lm-rank     | v3  | 2    | 2     | 5   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.580312 | current |
| POS  | lm-rank     | v4  | 3    | 3     | 1   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.620504 | current |
| POS  | lm-rank     | v5  | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.576452 | current |
| POS  | lm-rank     | v6  | 3    | 3     | 5   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.563856 | current |
| POS  | lm-rank     | v7  | 4    | 4     | 1   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.610426 | current |
| POS  | lm-rank     | v8  | 4    | 4     | 3   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.586282 | current |
| POS  | lm-rank     | v9  | 4    | 4     | 5   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.572337 | current |
| POS  | lm-rank     | v10 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | none               | F-Score | 0.549461 | current |
| POS  | lm-rank     | v11 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | inverse_parts      | F-Score | 0.555681 | current |
| POS  | lm-rank     | v12 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | inverse_sqrt_parts | F-Score | 0.566908 | current |
| POS  | lm-rank     | v13 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | inverse_log_parts  | F-Score | 0.537113 | current |
| POS  | lm-rank     | v14 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | none               | F-Score | 0.577517 | current |
| POS  | lm-rank     | v15 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | inverse_parts      | F-Score | 0.585044 | current |
| POS  | lm-rank     | v16 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | inverse_sqrt_parts | F-Score | 0.591720 | current |
| POS  | lm-rank     | v17 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | inverse_log_parts  | F-Score | 0.593328 | current |
| POS  | lm-rank     | v18 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | inverse_parts      | F-Score | 0.571601 | current |
| POS  | lm-rank     | v19 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | inverse_sqrt_parts | F-Score | 0.591173 | current |
| POS  | lm-rank     | v20 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | inverse_log_parts  | F-Score | 0.578678 | current |
| POS  | lm-rank     | v21 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | none               | F-Score | 0.577856 | current |
| POS  | lm-rank     | v22 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | inverse_parts      | F-Score | 0.575048 | current |
| POS  | lm-rank     | v23 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | inverse_sqrt_parts | F-Score | 0.581604 | current |
| POS  | lm-rank     | v24 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | inverse_log_parts  | F-Score | 0.572030 | current |
| POS  | lm-rank     | v25 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | none               | F-Score | 0.576826 | current |
| POS  | lm-rank     | v26 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | inverse_parts      | F-Score | 0.585848 | current |
| POS  | lm-rank     | v27 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | inverse_sqrt_parts | F-Score | 0.565807 | current |
| POS  | lm-rank     | v28 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | inverse_log_parts  | F-Score | 0.570471 | current |
| POS  | lm-skip     | v1  | 2    | 2     | 1   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.884458 | legacy  |
| POS  | lm-skip     | v2  | 2    | 2     | 3   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.848671 | legacy  |
| POS  | lm-skip     | v3  | 2    | 2     | 5   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.843252 | legacy  |
| POS  | lm-skip     | v4  | 3    | 3     | 1   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.882771 | legacy  |
| POS  | lm-skip     | v5  | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.852505 | legacy  |
| POS  | lm-skip     | v6  | 3    | 3     | 5   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.841513 | legacy  |
| POS  | lm-skip     | v7  | 4    | 4     | 1   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.882771 | legacy  |
| POS  | lm-skip     | v8  | 4    | 4     | 3   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.851227 | legacy  |
| POS  | lm-skip     | v9  | 4    | 4     | 5   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.842382 | legacy  |
| POS  | lm-subword  | v1  | 2    | 2     | 1   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.634551 | current |
| POS  | lm-subword  | v2  | 2    | 2     | 3   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.576961 | current |
| POS  | lm-subword  | v3  | 2    | 2     | 5   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.567349 | current |
| POS  | lm-subword  | v4  | 3    | 3     | 1   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.642950 | current |
| POS  | lm-subword  | v5  | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.601630 | current |
| POS  | lm-subword  | v6  | 3    | 3     | 5   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.561974 | current |
| POS  | lm-subword  | v7  | 4    | 4     | 1   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.629796 | current |
| POS  | lm-subword  | v8  | 4    | 4     | 3   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.599811 | current |
| POS  | lm-subword  | v9  | 4    | 4     | 5   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.574425 | current |
| POS  | lm-subword  | v10 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.581260 | current |
| POS  | lm-subword  | v11 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.591098 | current |
| POS  | lm-subword  | v12 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.578881 | current |
| POS  | lm-syllable | v1  | 2    | 2     | 1   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.633460 | current |
| POS  | lm-syllable | v2  | 2    | 2     | 3   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.572480 | current |
| POS  | lm-syllable | v3  | 2    | 2     | 5   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.586892 | current |
| POS  | lm-syllable | v4  | 3    | 3     | 1   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.629370 | current |
| POS  | lm-syllable | v5  | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.566346 | current |
| POS  | lm-syllable | v6  | 3    | 3     | 5   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.568616 | current |

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Table 1 — continued

| Task      | Method         | ID  | Win. | Slide | Top | Skip | Stem | Prune | $w_{\text{like}}$ | $w_{\text{prior}}$ | Penalty            | Metric  | Score    | Status  |
|-----------|----------------|-----|------|-------|-----|------|------|-------|-------------------|--------------------|--------------------|---------|----------|---------|
| POS       | lm-syllable    | v7  | 4    | 4     | 1   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.627398 | current |
| POS       | lm-syllable    | v8  | 4    | 4     | 3   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.581801 | current |
| POS       | lm-syllable    | v9  | 4    | 4     | 5   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.569290 | current |
| POS       | lm-word        | v1  | 10   | 7     | 10  | 20   | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.026706 | current |
| SENTIMENT | frequent-ngram | v1  | 2    | 2     | 1   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.580097 | current |
| SENTIMENT | frequent-ngram | v2  | 2    | 2     | 3   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.553256 | current |
| SENTIMENT | frequent-ngram | v3  | 2    | 2     | 5   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.585580 | current |
| SENTIMENT | frequent-ngram | v4  | 3    | 3     | 1   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.562360 | current |
| SENTIMENT | frequent-ngram | v5  | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.561976 | current |
| SENTIMENT | frequent-ngram | v6  | 3    | 3     | 5   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.556611 | current |
| SENTIMENT | frequent-ngram | v7  | 4    | 4     | 1   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.551513 | current |
| SENTIMENT | frequent-ngram | v8  | 4    | 4     | 3   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.574343 | current |
| SENTIMENT | frequent-ngram | v9  | 4    | 4     | 5   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.567692 | current |
| SENTIMENT | lm-lemma       | v1  | 2    | 2     | 1   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.637385 | current |
| SENTIMENT | lm-lemma       | v2  | 2    | 2     | 3   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.625718 | current |
| SENTIMENT | lm-lemma       | v3  | 2    | 2     | 5   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.606431 | current |
| SENTIMENT | lm-lemma       | v4  | 3    | 3     | 1   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.628652 | current |
| SENTIMENT | lm-lemma       | v5  | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.633384 | current |
| SENTIMENT | lm-lemma       | v6  | 3    | 3     | 5   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.620193 | current |
| SENTIMENT | lm-lemma       | v7  | 4    | 4     | 1   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.633145 | current |
| SENTIMENT | lm-lemma       | v8  | 4    | 4     | 3   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.617644 | current |
| SENTIMENT | lm-lemma       | v9  | 4    | 4     | 5   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.615423 | current |
| SENTIMENT | lm-lemma       | v10 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | none               | F-Score | 0.629875 | current |
| SENTIMENT | lm-lemma       | v11 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | inverse_parts      | F-Score | 0.635651 | current |
| SENTIMENT | lm-lemma       | v12 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | inverse_sqrt_parts | F-Score | 0.646935 | current |
| SENTIMENT | lm-lemma       | v13 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | inverse_log_parts  | F-Score | 0.622813 | current |
| SENTIMENT | lm-lemma       | v14 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | none               | F-Score | 0.611386 | current |
| SENTIMENT | lm-lemma       | v15 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | inverse_parts      | F-Score | 0.626651 | current |
| SENTIMENT | lm-lemma       | v16 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | inverse_sqrt_parts | F-Score | 0.625406 | current |
| SENTIMENT | lm-lemma       | v17 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | inverse_log_parts  | F-Score | 0.634553 | current |
| SENTIMENT | lm-lemma       | v18 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | inverse_parts      | F-Score | 0.626958 | current |
| SENTIMENT | lm-lemma       | v19 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | inverse_sqrt_parts | F-Score | 0.633720 | current |
| SENTIMENT | lm-lemma       | v20 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | inverse_log_parts  | F-Score | 0.620895 | current |
| SENTIMENT | lm-lemma       | v21 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | none               | F-Score | 0.635191 | current |
| SENTIMENT | lm-lemma       | v22 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | inverse_parts      | F-Score | 0.627702 | current |
| SENTIMENT | lm-lemma       | v23 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | inverse_sqrt_parts | F-Score | 0.626574 | current |
| SENTIMENT | lm-lemma       | v24 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | inverse_log_parts  | F-Score | 0.619633 | current |
| SENTIMENT | lm-lemma       | v25 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | none               | F-Score | 0.615608 | current |
| SENTIMENT | lm-lemma       | v26 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | inverse_parts      | F-Score | 0.636217 | current |
| SENTIMENT | lm-lemma       | v27 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | inverse_sqrt_parts | F-Score | 0.638458 | current |
| SENTIMENT | lm-lemma       | v28 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | inverse_log_parts  | F-Score | 0.624693 | current |
| SENTIMENT | lm-rank        | v1  | 2    | 2     | 1   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.610352 | current |
| SENTIMENT | lm-rank        | v2  | 2    | 2     | 3   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.603944 | current |
| SENTIMENT | lm-rank        | v3  | 2    | 2     | 5   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.607960 | current |
| SENTIMENT | lm-rank        | v4  | 3    | 3     | 1   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.609935 | current |
| SENTIMENT | lm-rank        | v5  | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.592764 | current |
| SENTIMENT | lm-rank        | v6  | 3    | 3     | 5   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.599763 | current |
| SENTIMENT | lm-rank        | v7  | 4    | 4     | 1   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.625708 | current |
| SENTIMENT | lm-rank        | v8  | 4    | 4     | 3   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.614066 | current |
| SENTIMENT | lm-rank        | v9  | 4    | 4     | 5   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.615280 | current |
| SENTIMENT | lm-rank        | v10 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | none               | F-Score | 0.608788 | current |
| SENTIMENT | lm-rank        | v11 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | inverse_parts      | F-Score | 0.603412 | current |
| SENTIMENT | lm-rank        | v12 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | inverse_sqrt_parts | F-Score | 0.618265 | current |
| SENTIMENT | lm-rank        | v13 | 3    | 3     | 3   | 3    | 7    | 100   | 0.0               | 1.0                | inverse_log_parts  | F-Score | 0.614529 | current |
| SENTIMENT | lm-rank        | v14 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | none               | F-Score | 0.612632 | current |
| SENTIMENT | lm-rank        | v15 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | inverse_parts      | F-Score | 0.612582 | current |
| SENTIMENT | lm-rank        | v16 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | inverse_sqrt_parts | F-Score | 0.613280 | current |
| SENTIMENT | lm-rank        | v17 | 3    | 3     | 3   | 3    | 7    | 100   | 0.05              | 0.95               | inverse_log_parts  | F-Score | 0.625213 | current |
| SENTIMENT | lm-rank        | v18 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | inverse_parts      | F-Score | 0.612742 | current |
| SENTIMENT | lm-rank        | v19 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | inverse_sqrt_parts | F-Score | 0.617745 | current |
| SENTIMENT | lm-rank        | v20 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | inverse_log_parts  | F-Score | 0.614958 | current |
| SENTIMENT | lm-rank        | v21 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | none               | F-Score | 0.605059 | current |
| SENTIMENT | lm-rank        | v22 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | inverse_parts      | F-Score | 0.577309 | current |
| SENTIMENT | lm-rank        | v23 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | inverse_sqrt_parts | F-Score | 0.612739 | current |

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Table 1 — continued

| Task      | Method      | ID  | Win. | Slide | Top | Skip | Stem | Prune | $w_{\text{like}}$ | $w_{\text{prior}}$ | Penalty            | Metric  | Score    | Status  |
|-----------|-------------|-----|------|-------|-----|------|------|-------|-------------------|--------------------|--------------------|---------|----------|---------|
| SENTIMENT | lm-rank     | v24 | 3    | 3     | 3   | 3    | 7    | 100   | 0.3               | 0.7                | inverse_log_parts  | F-Score | 0.626831 | current |
| SENTIMENT | lm-rank     | v25 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | none               | F-Score | 0.605226 | current |
| SENTIMENT | lm-rank     | v26 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | inverse_parts      | F-Score | 0.599042 | current |
| SENTIMENT | lm-rank     | v27 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | inverse_sqrt_parts | F-Score | 0.621683 | current |
| SENTIMENT | lm-rank     | v28 | 3    | 3     | 3   | 3    | 7    | 100   | 0.5               | 0.5                | inverse_log_parts  | F-Score | 0.607730 | current |
| SENTIMENT | lm-skip     | v1  | 2    | 2     | 1   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.664706 | current |
| SENTIMENT | lm-skip     | v2  | 2    | 2     | 3   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.611257 | current |
| SENTIMENT | lm-skip     | v3  | 2    | 2     | 5   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.600008 | current |
| SENTIMENT | lm-skip     | v4  | 3    | 3     | 1   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.678592 | current |
| SENTIMENT | lm-skip     | v5  | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.622701 | current |
| SENTIMENT | lm-skip     | v6  | 3    | 3     | 5   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.617793 | current |
| SENTIMENT | lm-skip     | v7  | 4    | 4     | 1   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.648283 | current |
| SENTIMENT | lm-skip     | v8  | 4    | 4     | 3   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.620893 | current |
| SENTIMENT | lm-skip     | v9  | 4    | 4     | 5   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.621868 | current |
| SENTIMENT | lm-subword  | v1  | 2    | 2     | 1   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.653932 | current |
| SENTIMENT | lm-subword  | v2  | 2    | 2     | 3   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.622011 | current |
| SENTIMENT | lm-subword  | v3  | 2    | 2     | 5   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.599383 | current |
| SENTIMENT | lm-subword  | v4  | 3    | 3     | 1   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.669665 | current |
| SENTIMENT | lm-subword  | v5  | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.606105 | current |
| SENTIMENT | lm-subword  | v6  | 3    | 3     | 5   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.609024 | current |
| SENTIMENT | lm-subword  | v7  | 4    | 4     | 1   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.661130 | current |
| SENTIMENT | lm-subword  | v8  | 4    | 4     | 3   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.619042 | current |
| SENTIMENT | lm-subword  | v9  | 4    | 4     | 5   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.612283 | current |
| SENTIMENT | lm-subword  | v10 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.626043 | current |
| SENTIMENT | lm-subword  | v11 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.610327 | current |
| SENTIMENT | lm-subword  | v12 | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.609594 | current |
| SENTIMENT | lm-syllable | v1  | 2    | 2     | 1   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.661312 | current |
| SENTIMENT | lm-syllable | v2  | 2    | 2     | 3   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.635142 | current |
| SENTIMENT | lm-syllable | v3  | 2    | 2     | 5   | 2    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.604265 | current |
| SENTIMENT | lm-syllable | v4  | 3    | 3     | 1   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.644224 | current |
| SENTIMENT | lm-syllable | v5  | 3    | 3     | 3   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.613723 | current |
| SENTIMENT | lm-syllable | v6  | 3    | 3     | 5   | 3    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.616151 | current |
| SENTIMENT | lm-syllable | v7  | 4    | 4     | 1   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.656830 | current |
| SENTIMENT | lm-syllable | v8  | 4    | 4     | 3   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.620066 | current |
| SENTIMENT | lm-syllable | v9  | 4    | 4     | 5   | 4    | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.612592 | current |
| SENTIMENT | lm-word     | v1  | 10   | 7     | 10  | 20   | 7    | 100   | 0.15              | 0.85               | none               | F-Score | 0.651535 | current |